

MH370 stringent multi-debris drift diagnostic - separated recovered-object and non-recovery populations

Uncalibrated family-specific diagnostic; currents=cmems_glorys12_reanalysis | isotope=enabled as a neutral-or-penalizing screen conditional on represented flaperon arrival

Estimate at a glance

CURRENT FAMILY

cmems_glorys12_reanalysis

PEAK POSITION

-32.638 deg , 95.768 deg

RECOVERY EVIDENCE

9 objects in 9 independent episodes

MEAN PROPOSAL TERMINATION FRACTION

72.6%

PEAK SOURCE CELL

arc-156

WEIGHTED TRAJECTORIES

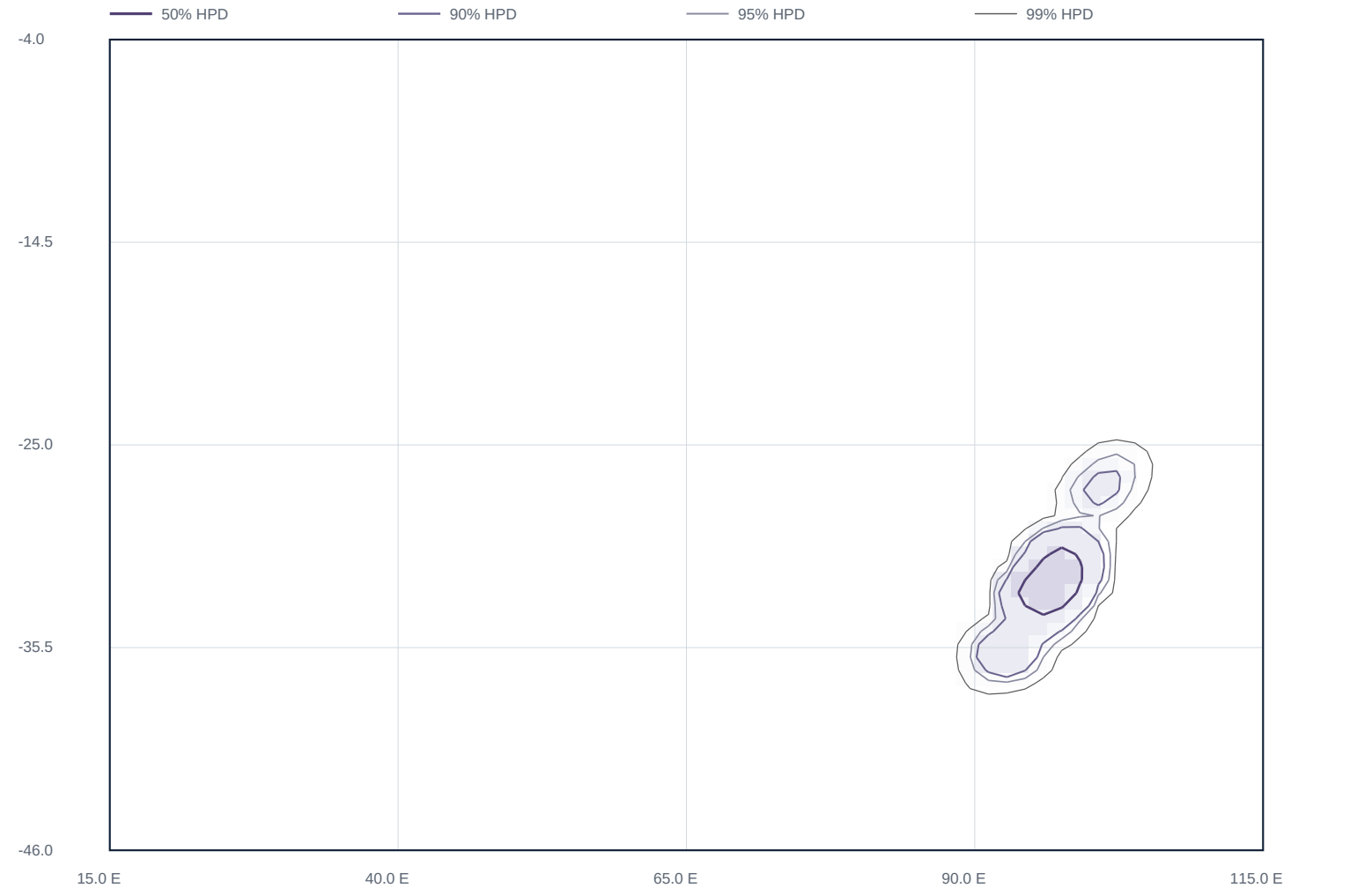
1024 x 77 cells x 11 transport ensembles (9 event-targeted r

MEAN IMPORTANCE-WEIGHTED FIELD-TERMINATION FRACTION

73.2%

ARRIVAL SAMPLER

AdaptiveSmc

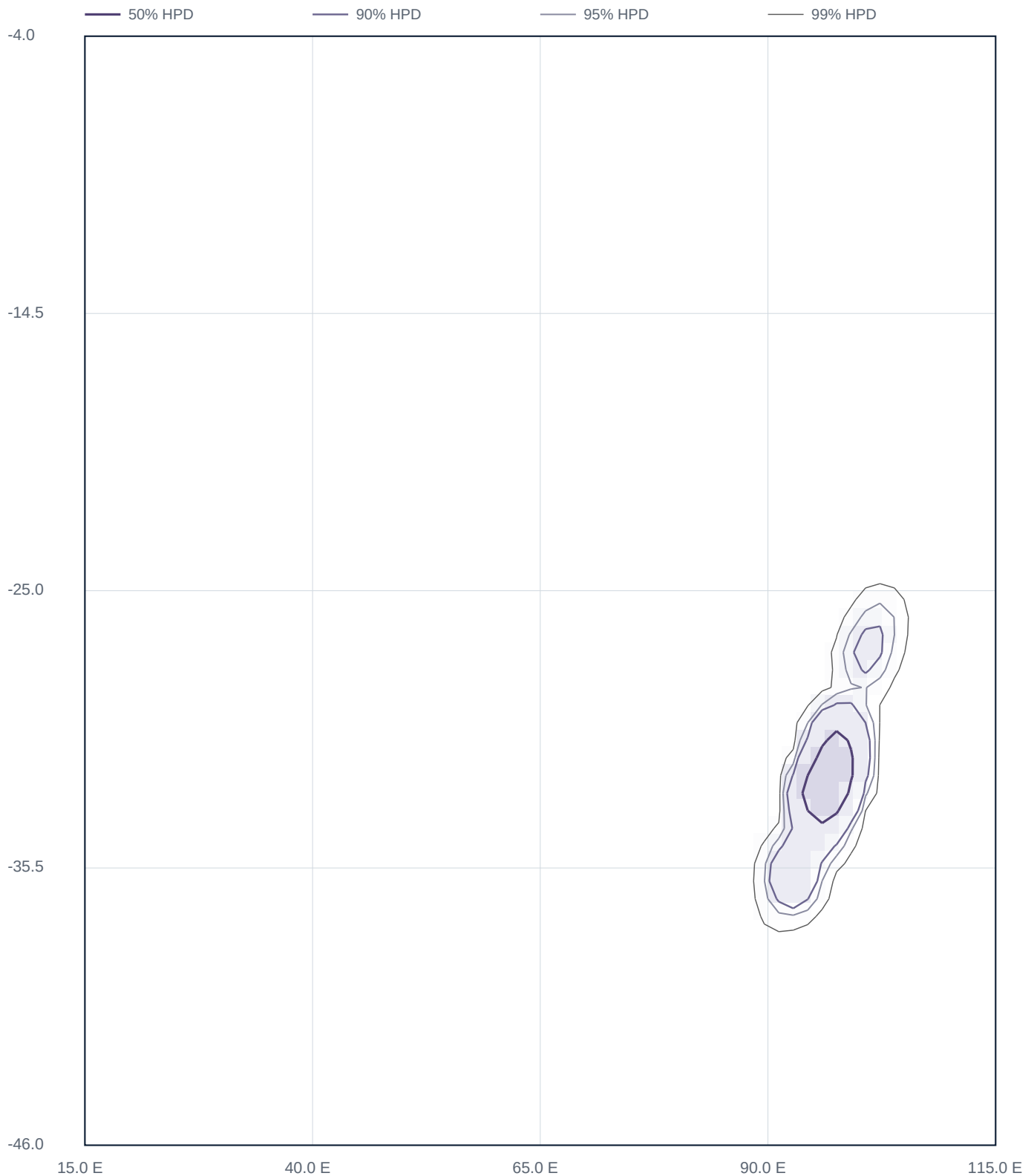


Probability is conditional on the named motion, SATCOM, and evidence configuration.
HPD contours use a fixed 5-cell display kernel; estimator weights and numerical summaries are unchanged.

Posterior geography

Highest-density regions are computed from the weighted particle population.

HPD contours use a fixed 5-cell display kernel; estimator weights and numerical summaries are unchanged.

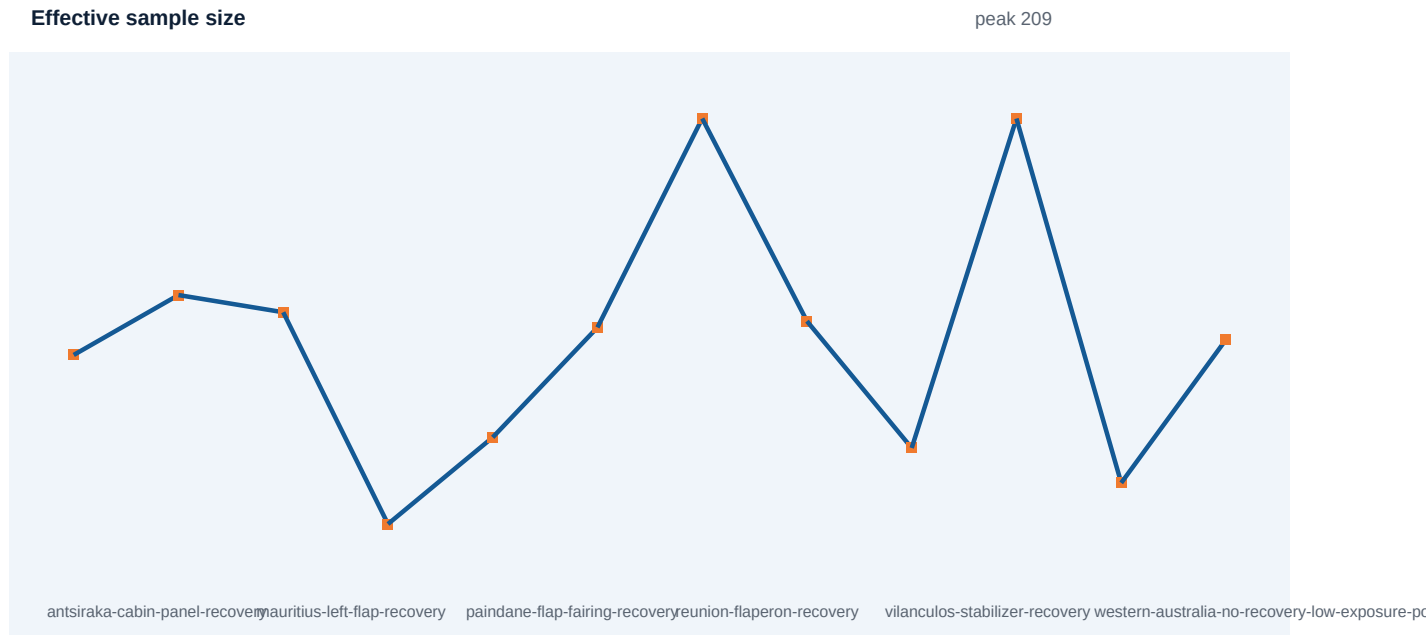


Interpretation

The map is a probability density over the modelled impact state. It is not a searched-area elimination map, and structural model alternatives must be compared separately.

Numerical behaviour and conditions

Adaptive likelihood tempering, particle diversity, and declared evidence conditions.



Evidence conditions

- Current family selected explicitly: cmems_glorys12_reanalysis
- Object-motion families selected explicitly: measured-flaperon-response, low-exposure-exterior, uncertain-interior-panels-low-exposure-comparator, unobserved-low-exposure-non-recovery-population, unobserved-high-windage-non-recovery-population
- Current persistent-land corners renormalized=true; intermittent missing values remain fatal
- Barnacle isotope: enabled as a neutral-or-penalizing screen conditional on represented flaperon arrival
- distinct direct 10 m windage: measured-flaperon-response=0.000% + 0.100 m/s at -20.0 degrees, unobserved-high-windage-non-recovery-population=3.000% + 0.000 m/s at 0.0 degrees
- distinct Stokes velocity field enabled; responses measured-flaperon-response=100%, low-exposure-exterior=100%, uncertain-interior-panels-low-exposure-comparator=100%, unobserved-low-exposure-non-recovery-population=100%, unobserved-high-windage-non-recovery-population=0%; finite coastal corners renormalized=true
- explicit land mask enabled; coastal beaching disabled because no beaching-rate field was supplied
- Current interpolation across timestamp gaps greater than 36.0 h is outside-time support

Material limitations

- Diagnostic source-compatibility result, not a calibrated final MH370 posterior.
- Current families are reported separately and are never averaged using subjective weights.
- A2-G1 covers only the final months; the screen is neutral for any compatible chronology and never density-ranks compatible paths.
- No field-unavailable trajectory is relabelled as beaching; termination fractions and reasons remain in the JSON artifact.
- GLORYS12V1 is assimilative; the available drifter replay is not guaranteed held out.
- Daily means omit subdaily current variability represented by the native-HYCOM three-hour analyses.
- Land and any missing ocean value remain the packed missing sentinel; no missing value is converted to zero current.
- Stokes drift, direct windage, and coastal beaching are not included in this current field and require distinct explicit inputs.